

# Shota Ikari

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## Education

### The University of Tokyo, Japan

Apr 2025 – Present

Master's Program (Expected Mar 2027), Department of Information Physics and Computing,  
Graduate School of Information Science and Technology  
Computing Systems Research Laboratory

- GPA: 4.0/4.0
- Research: Quantum Computer Architecture.

### Kyoto University, Japan

Apr 2021 – Mar 2025

Bachelor of Engineering, Department of Electrical and Electronic Engineering, Faculty of Engineering  
Photonic Quantum Information Laboratory

- GPA: 3.88/4.3
- Bachelor's thesis:  
*Quantum Walks with Periodic Boundary Conditions for Observation of Non-Hermitian Physics*

## International Conference Proceedings

### Circuit-Level Implementation and Evaluation of Logical SH Gates in the Surface Code

2026

Shota Ikari, Yuga Hirai, Yosuke Ueno, Yasunari Suzuki, and Hiroshi Nakamura

IEEE International Conference on Quantum Computing and Engineering (QCE). To appear; poster.

## Preprints

### Do Not Let CNOTs Overwhelm the Decoder: Scheduling Transversal Gates for Fast FTQC

2026

Shota Ikari, Yuga Hirai, Yasunari Suzuki, Hiroshi Nakamura, and Yosuke Ueno

arXiv:2608.11719 [quant-ph]

### A $2d \times d \times d$ Spacetime Volume Implementation of a Logical S Gate in the Surface Code

2026

Yuga Hirai, Shota Ikari, Yosuke Ueno, and Yasunari Suzuki

arXiv:2604.13632 [quant-ph]

### No More Hooks in the Surface Code: Distance-Preserving Syndrome Extraction for Arbitrary Layouts at Minimum Depth

2026

Yuga Hirai, Shota Ikari, Yosuke Ueno, and Yasunari Suzuki

arXiv:2603.01628 [quant-ph]

## Scholarships and Grants

### Kyoto University CF Project Scholarship

Apr 2023 – Mar 2025

Issued by Kyoto University

- Stipend: JPY 50,000 / month
- Awarded based on academic achievement

### Scholarship for Japanese University Students

Jul 2024

Issued by Keyence Foundation

- One-time grant: JPY 300,000
- Selected based on application materials and an essay

## Work Experience

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**Part-time Research Engineer**, Preferred Networks, Inc. May 2025 – Present

- Develop and implement statistical models for weather data using PyTorch.
- Optimize GPU training performance and data processing pipelines for large-scale datasets.

**Software Engineer Intern**, Recruit Co., Ltd. Jan 2026 – Feb 2026

- Built a full-stack LLM application integrating RAG and MCP servers.
- Deployed the application on AWS and configured a CI/CD pipeline.

**Teaching Assistant**, The University of Tokyo Apr 2025 – Aug 2026

- Assisted students with experiments involving microcontrollers and FPGAs.

**Part-time Software Engineer**, DATAGRID Mar 2022 – Aug 2024

- Investigated computer vision and machine learning methods for image processing and generation.
- Formulated optimization models for transportation scheduling using Python and OR-Tools.